DPO and GRPO

RL without a critic, and preferences without RL

Carlos Cotrini

ETH Zurich

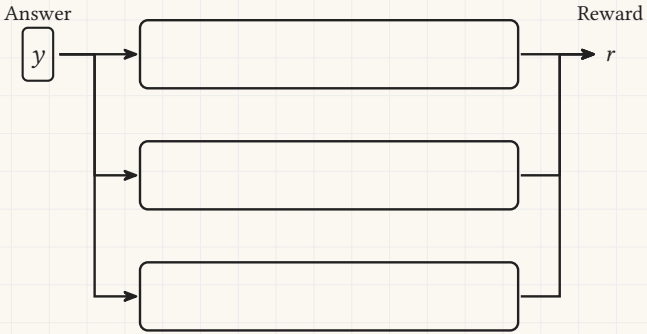
20 October 2026

1

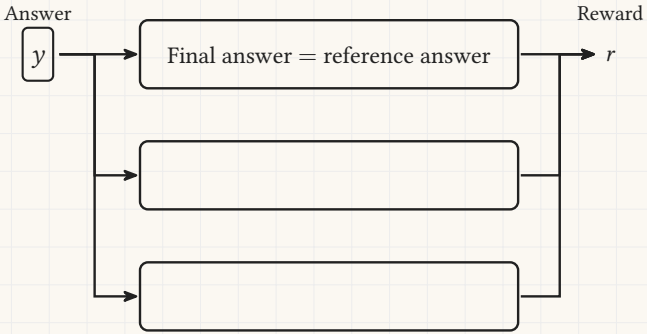
Section 1

Verifiable rewards

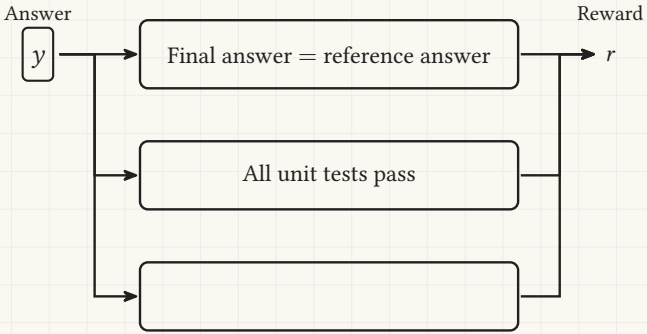
Rewards a program can check



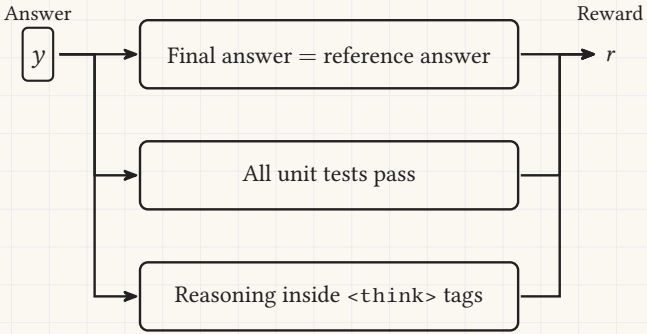
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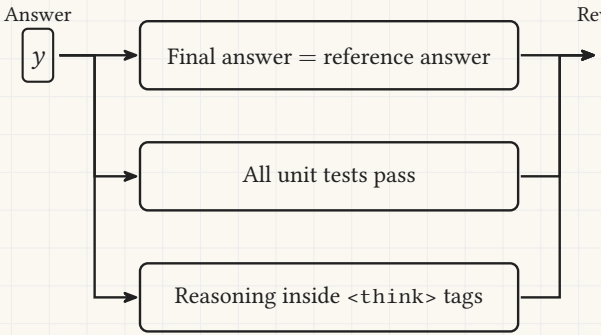
Rewards a program can check



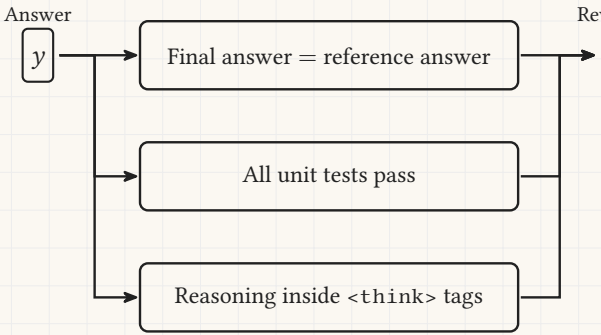
Rewards a program can check



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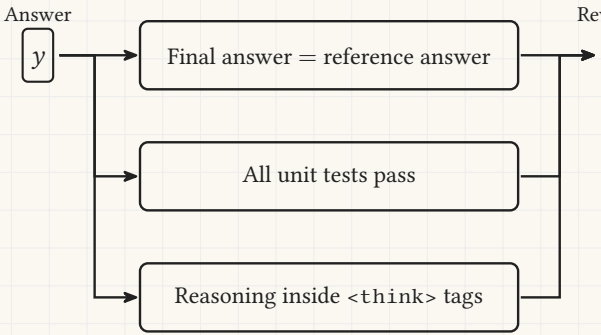


Rewards a program can check



$r \in \{0, 1\}$	Liu et al. 2025
$\in \{-1, 1\}$	Yu et al. 2025
$\in \{0, 10\}$	Lambert et al. 2025
$=$ accuracy $+$ format	Guo et al. 2025

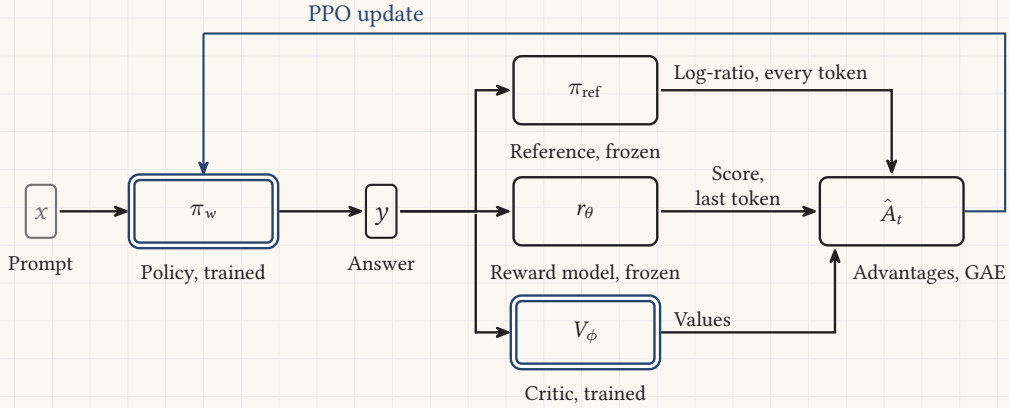
Rewards a program can check



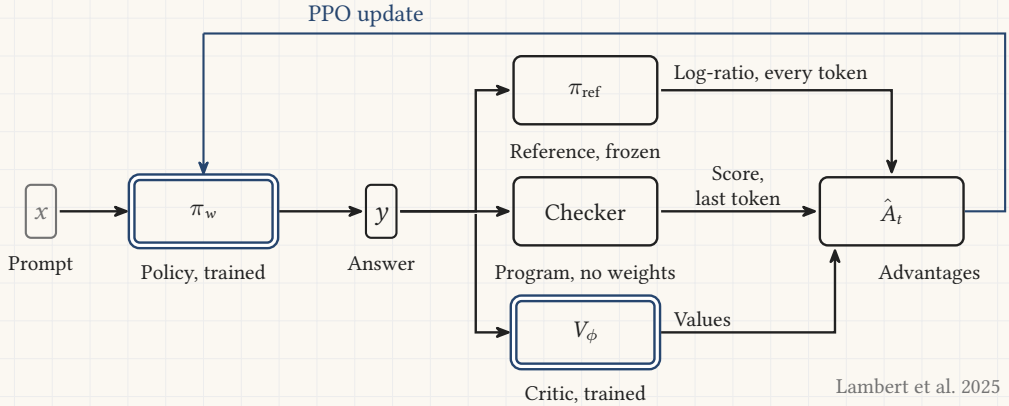
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$\in \{0, 10\}$	Lambert et al. 2025
$=$ accuracy $+$ format	Guo et al. 2025

DeepSeek-R1: learned reward models get hacked; retraining costs compute.

The same loop with a checker

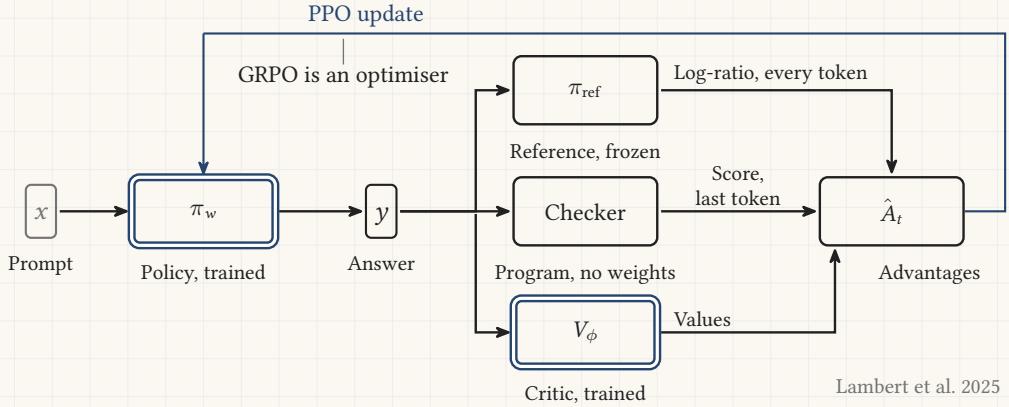


The same loop with a checker



Lambert et al. 2025

The same loop with a checker



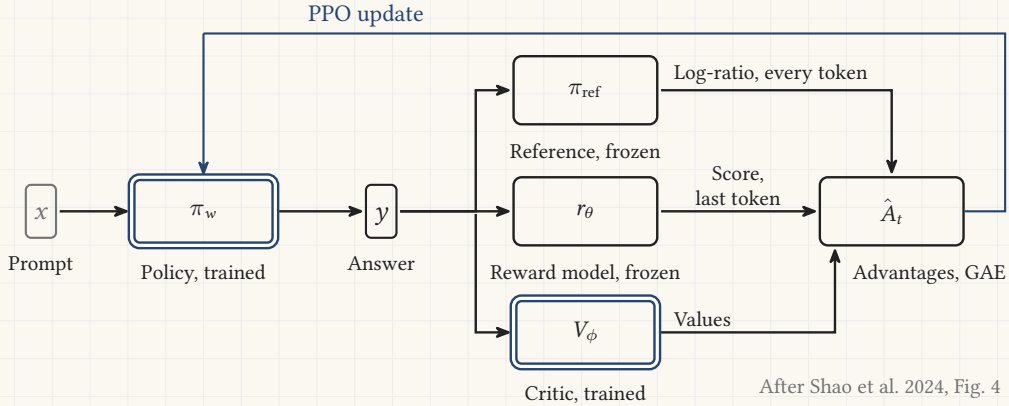
Lambert et al. 2025

2

Section 2

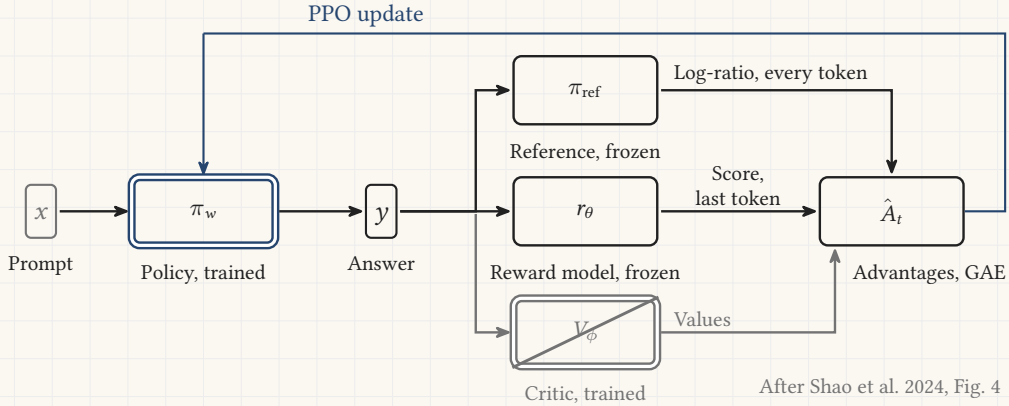
GRPO

From PPO to GRPO



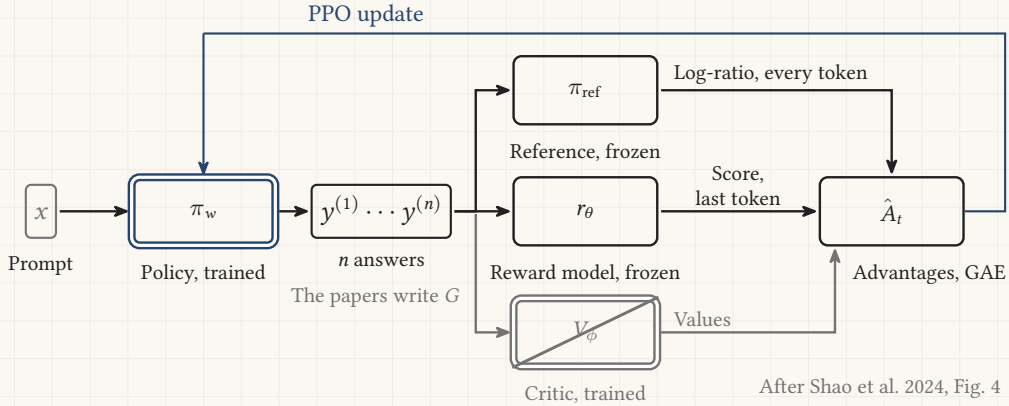
After Shao et al. 2024, Fig. 4

From PPO to GRPO



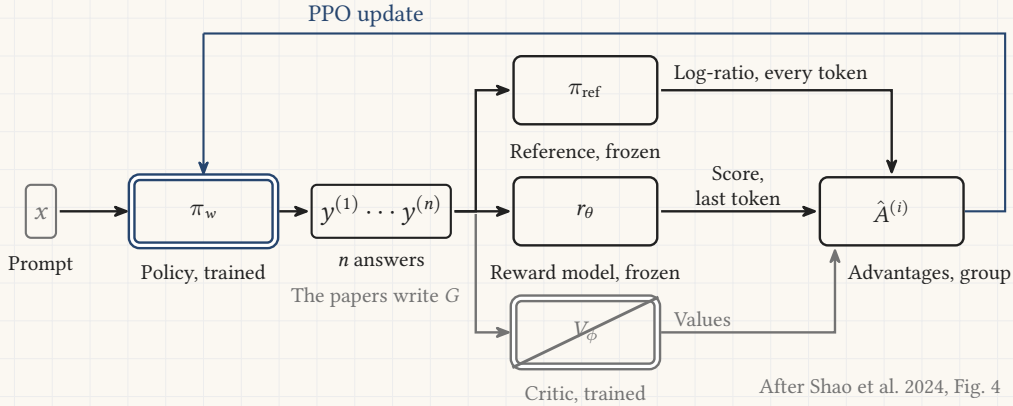
After Shao et al. 2024, Fig. 4

From PPO to GRPO

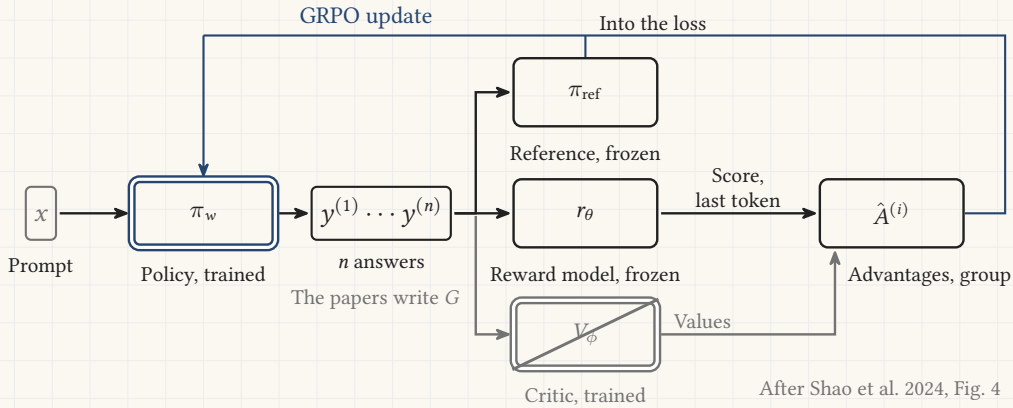


After Shao et al. 2024, Fig. 4

From PPO to GRPO

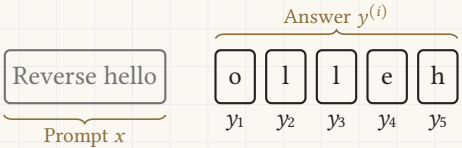


From PPO to GRPO

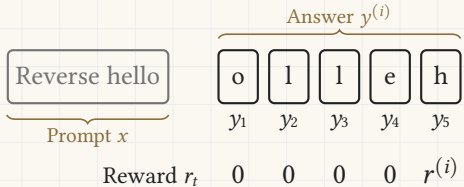


After Shao et al. 2024, Fig. 4

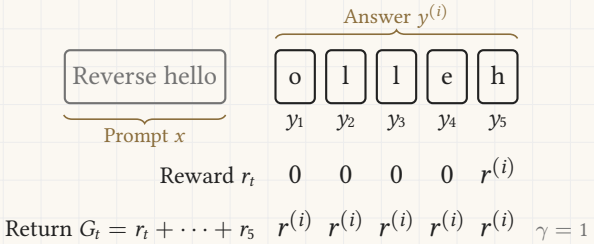
Every token's return is the answer's reward



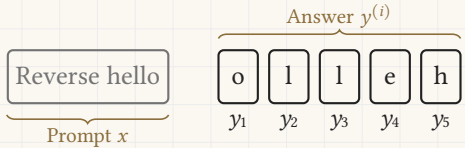
Every token's return is the answer's reward



Every token's return is the answer's reward



Every token's return is the answer's reward



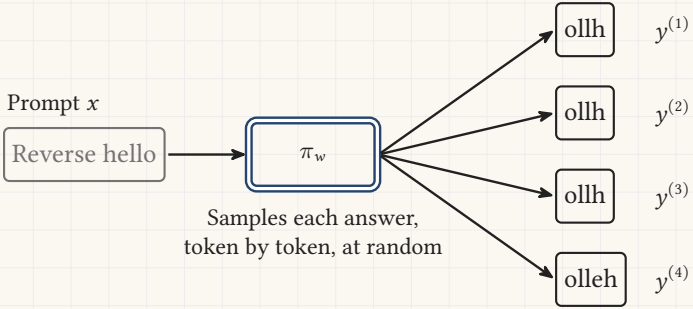
Reward r_t	0	0	0	0	$r^{(i)}$		Every token of $y^{(i)}$: multiplier
Return $G_t = r_t + \cdots + r_5$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$r^{(i)}$	$\gamma = 1$	$G_t - \text{baseline} = r^{(i)} - \text{baseline}$

The group: n answers to the same prompt



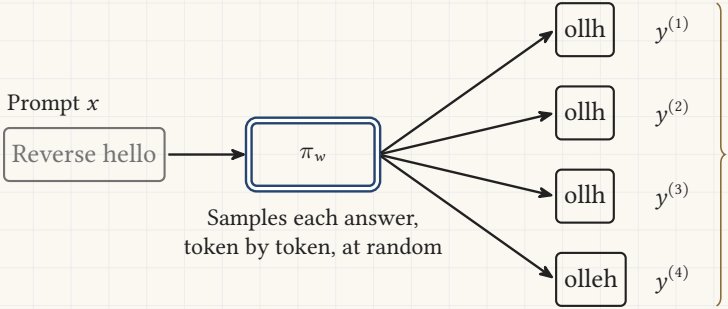
Welleck 2026, CMU CS11-711, lecture 17

The group: n answers to the same prompt



Welleck 2026, CMU CS11-711, lecture 17

The group: n answers to the same prompt



GRPO: Group Relative Policy Optimization

The group of x: its n answers

DeepSeekMath: $n = 64$

Shao et al. 2024, Sec. 4.2

Welleck 2026, CMU CS11-711, lecture 17

The group average is the baseline

Prompt x Reverse hello

Welleck 2026, CMU CS11-711, lecture 17

Answer $y^{(i)}$ | ollh | ollh | ollh | olleh | $n = 4$, sampled from π_w

The group average is the baseline

Prompt x Reverse hello

Welleck 2026, CMU CS11-711, lecture 17

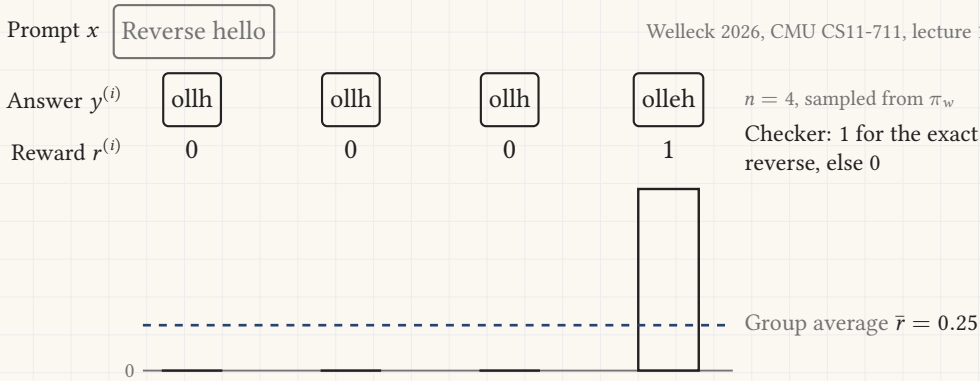
Answer $y^{(i)}$	ollh	ollh	ollh	olleh
Reward $r^{(i)}$	0	0	0	1

$n = 4$, sampled from π_w

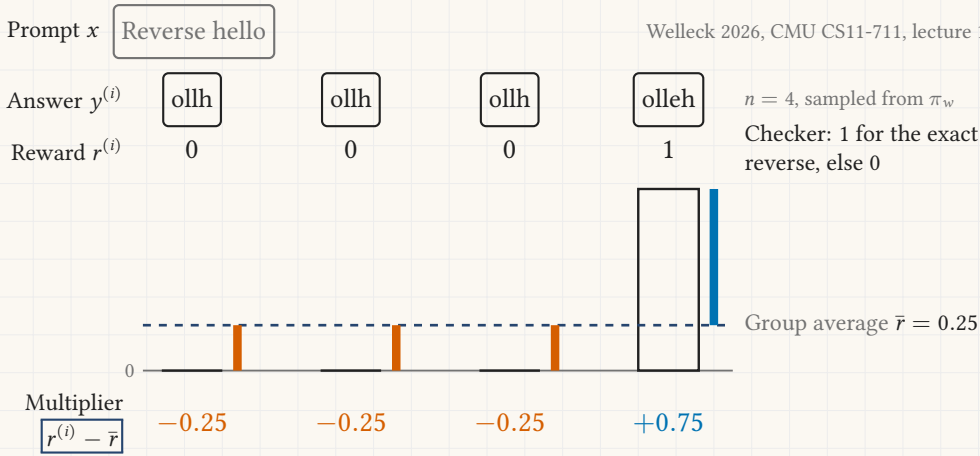
Checker: 1 for the exact reverse, else 0



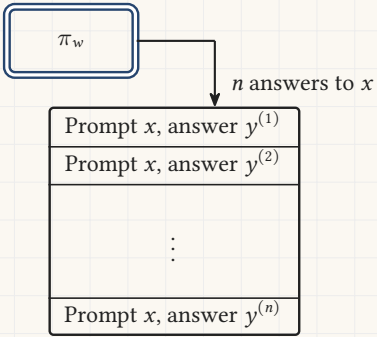
The group average is the baseline



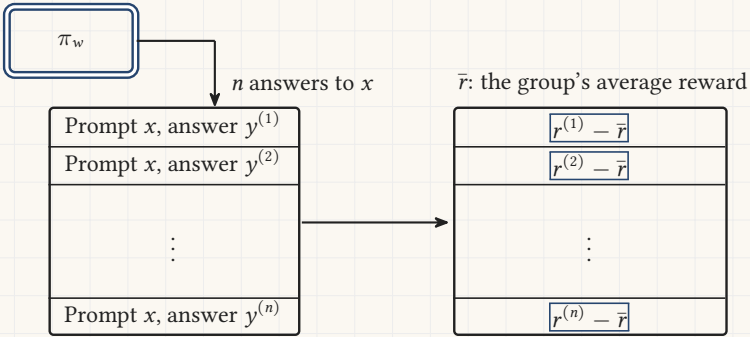
The group average is the baseline



The same loop for GRPO

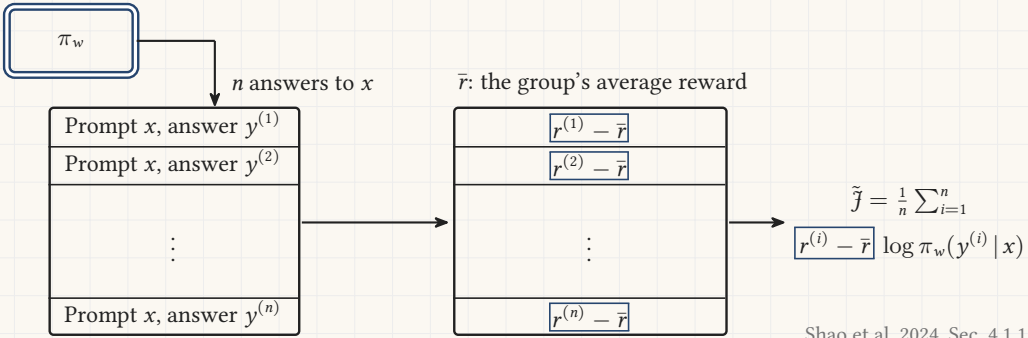


The same loop for GRPO



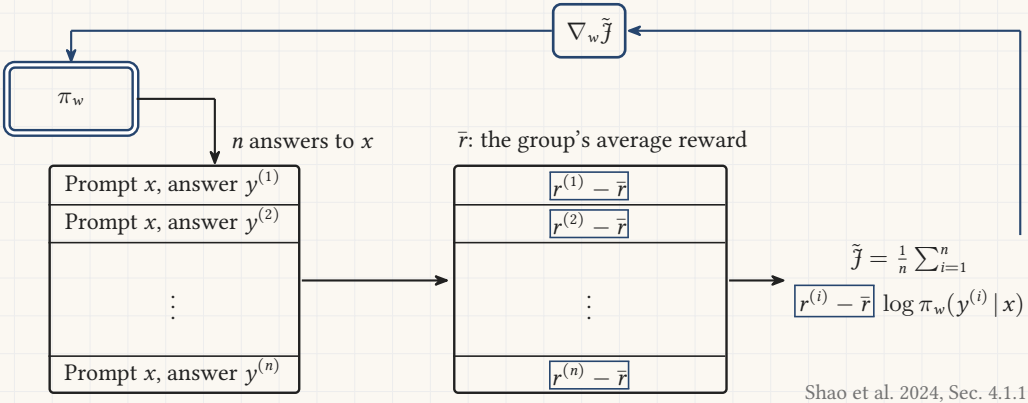
Shao et al. 2024, Sec. 4.1.1

The same loop for GRPO



Shao et al. 2024, Sec. 4.1.1

The same loop for GRPO



Shao et al. 2024, Sec. 4.1.1

3

Section 3, optional

GRPO in detail

Leaving the answer out only rescales the multiplier

Morning's rule: no baseline may see the answer it scores

$\bar{r}$ includes it; RLOO uses $\bar{r}_{-i}$, the others' average



Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

Leaving the answer out only rescales the multiplier

Morning's rule: no baseline may see the answer it scores

$\bar{r}$ includes it; RLOO uses $\bar{r}_{-i}$, the others' average

Answer $y^{(i)}$	ollh	ollh	ollh	olleh
RLOO $r^{(i)} - \bar{r}_{-i}$	-0.33	-0.33	-0.33	$+1$
GRPO $r^{(i)} - \bar{r}$	-0.25	-0.25	-0.25	$+0.75$

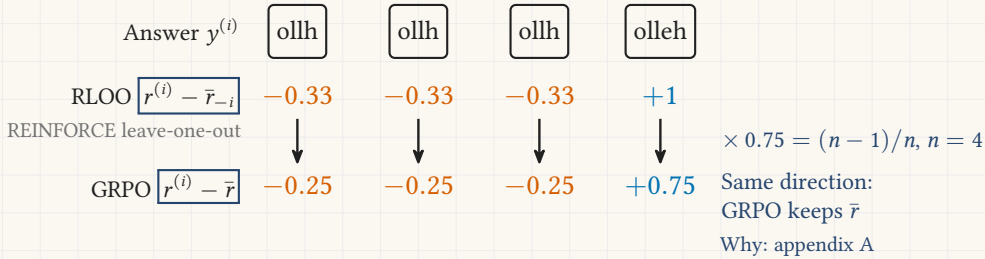
REINFORCE leave-one-out

Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

Leaving the answer out only rescales the multiplier

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Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The GRPO objective

Shao et al. 2024, eq. (3)

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\substack{\text{Over answers, then tokens:} \\ \text{divided by the answer's length}}} \Big[\qquad \Big]$$

The GRPO objective

Shao et al. 2024, eq. (3)

$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\substack{\text{Over answers, then tokens:} \\ \text{divided by the answer's length}}} \Bigg[\underbrace{\min\Big(\rho_t \overbrace{\boxed{\hat{A}^{(i)}}}^{\substack{\text{Group advantage:} \\ \text{divided by the spread}}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \boxed{\hat{A}^{(i)}}\Big)}_{\text{PPO clipping}} \Bigg]$$

The GRPO objective

Shao et al. 2024, eq. (3)

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The GRPO objective

Shao et al. 2024, eq. (3)

One update per batch:
$\rho_t = 1 \Rightarrow$ policy gradient

Group advantage:
divided by the spread

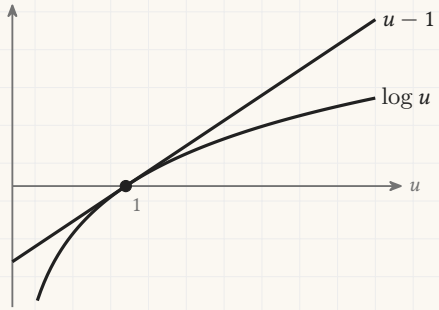
$$L(w) = \underbrace{\text{average}_i \ \text{average}_t}_{\text{Over answers, then tokens: divided by the answer's length}} \Big[\underbrace{\min\big(\rho_t \, \boxed{\hat{A}^{(i)}}, \ \text{clip}(\rho_t, \, 1-\epsilon, \, 1+\epsilon) \, \boxed{\hat{A}^{(i)}}\big)}_{\text{PPO clipping}} - \underbrace{\beta \, k_t}_{\text{KL estimate, weight } \beta} \Big]$$

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_{\text{w}}, \text{ per token}$$

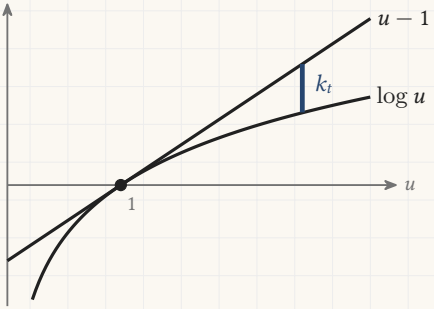
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The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_w, \text{ per token}$$

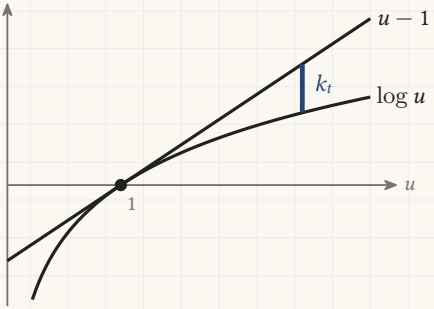


Never negative; unbiased for tokens from π_w

Schulman 2020

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_w, \text{ per token}$$



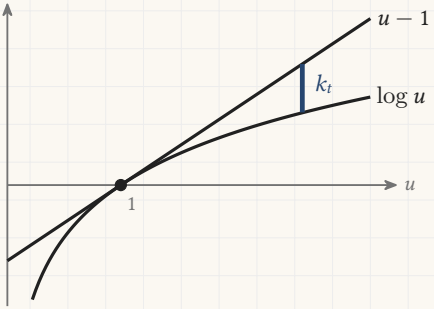
Never negative; unbiased for tokens from π_w

Schulman 2020

Weight β	
DeepSeekMath	0.04
DeepSeek-R1-Zero	0.001 New π_{ref} every 400 steps
Two later variants	0

The KL estimate in the loss

$$k_t = (u - 1) - \log u \quad u = \pi_{\text{ref}}/\pi_w, \text{ per token}$$



Never negative; unbiased for tokens from π_w

Schulman 2020

Weight β

DeepSeekMath	0.04
DeepSeek-R1-Zero	0.001
	New π_{ref} every 400 steps
Two later variants	0

In the reward: may penalise length

Guo et al. 2025

Two later variants: Dr. GRPO and DAPO

Dr. GRPO

GRPO Done Right

Problem	GRPO's two divisions bias it
Change	Drop both
Gain	Same accuracy, shorter wrong answers

Liu et al. 2025

Two later variants: Dr. GRPO and DAPO

	Dr. GRPO GRPO Done Right	DAPO
Problem	GRPO's two divisions bias it	Agreeing groups teach nothing
Change	Drop both	Drop them, refill the batch
Gain	Same accuracy, shorter wrong answers	AIME 2024: 42 → 50
	Liu et al. 2025	Yu et al. 2025, Table 1

Dividing by the spread removes the reward's units

Rewards		Minus $\bar{r}$ (ollh, olleh)
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$
$r \in \{-1, 1\}$	Yu et al. 2025	$-0.50,\ +1.50$

Dividing by the spread removes the reward's units

DeepSeekMath
Shao et al. 2024, Sec. 4.1.2

$$\hat{A}^{(i)} = \frac{r^{(i)} - \bar{r}}{\underbrace{\text{std}}}$$

Spread of the group's rewards

Rewards		Minus $\bar{r}$ (ollh, olleh)
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$
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Dividing by the spread removes the reward's units

DeepSeekMath

Shao et al. 2024, Sec. 4.1.2

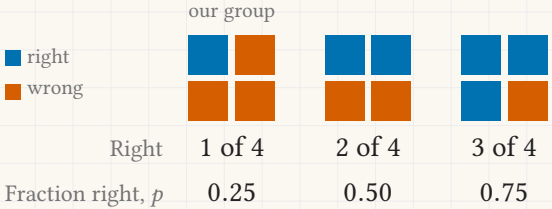
$$\hat{A}^{(i)} = \frac{r^{(i)} - \bar{r}}{\underbrace{\text{std}}_{\text{Spread of the group's rewards}}}$$

Every group gets spread 1:
one step size and one β for any reward

Rewards		Minus $\bar{r}$ (ollh, olleh)	Divided by std: $\hat{A}^{(i)}$	
$r \in \{0, 1\}$	Liu et al. 2025	$-0.25,\ +0.75$	$-0.577,\ +1.732$	
$r \in \{0, 10\}$	Lambert et al. 2025	$-2.50,\ +7.50$	$-0.577,\ +1.732$	
$r \in \{-1, 1\}$	Yu et al. 2025	$-0.50,\ +1.50$	$-0.577,\ +1.732$	std over n

The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0

our group

right
wrong



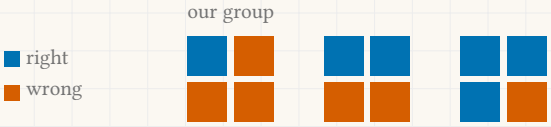
Right	1 of 4	2 of 4	3 of 4
Fraction right, p	0.25	0.50	0.75
Spread, std	0.433	0.500	0.433

$$\text{std} = \sqrt{p\,(1-p)}$$

rewards 1 or 0, std over n

The spread depends only on the fraction right

Three prompts, $n = 4$ answers each, rewards 1 or 0



Right	1 of 4	2 of 4	3 of 4
Fraction right, p	0.25	0.50	0.75
Spread, std	0.433	0.500	0.433

$$\text{std} = \sqrt{p\,(1-p)}$$

rewards 1 or 0, std over n

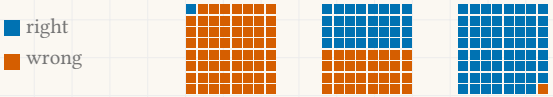
Set by p alone,
and p is how hard the prompt is

few right: hard; most right: easy

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2

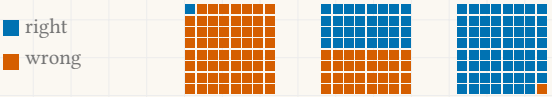


Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2

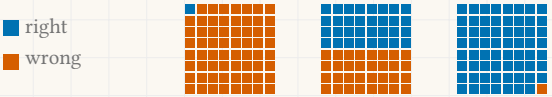


Right	1 of 64	32 of 64	63 of 64
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Spread, std over n	0.124	0.500	0.124

Dividing favours prompts whose answers nearly agree

DeepSeekMath's $n = 64$ answers per prompt; a checker's 1 or 0

Shao et al. 2024, Sec. 4.2



Right	1 of 64	32 of 64	63 of 64
Fraction right, p	0.016	0.500	0.984
Spread, std over n	0.124	0.500	0.124
Each multiplier $\times$ 1/std	$\times$ 8.06	$\times$ 2.00	$\times$ 8.06

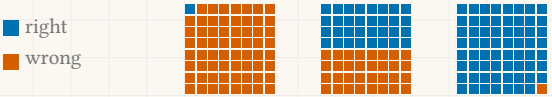
A difficulty bias

Liu et al. 2025, Sec. 3.1

Dividing favours prompts whose answers nearly agree

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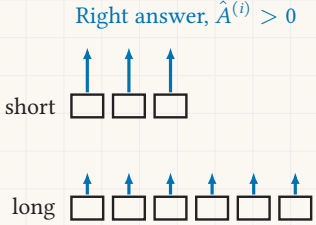
Rewards 1 or 0: no units to remove;
Dr. GRPO drops the division

Dividing by the length favours long wrong answers

GRPO: each token of $y^{(i)}$ gets $\hat{A}^{(i)}/|y^{(i)}|$

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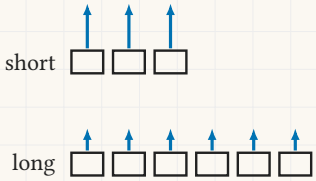
Short ones gain more per token

Dividing by the length favours long wrong answers

GRPO: each token of $y^{(i)}$ gets $\hat{A}^{(i)}/|y^{(i)}|$

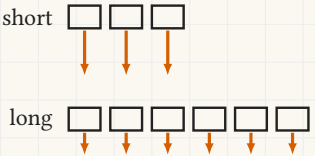
Liu et al. 2025, Sec. 3.1

Right answer, $\hat{A}^{(i)} > 0$



Short ones gain more per token

Wrong answer, $\hat{A}^{(i)} < 0$



Long ones lose less per token:
wrong answers grow longer

Dr. GRPO drops both divisions

Each token of $y^{(i)}$ gets, in GRPO:

$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{\text{std}} \times \frac{1}{\left|y^{(i)}\right|}$$

Dr. GRPO drops both divisions

Each token of $y^{(i)}$ gets, in GRPO:

$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{\text{std}} \times \frac{1}{|y^{(i)}|}$$

Difficulty bias Length bias

Liu et al. 2025, Sec. 3.2

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Difficulty bias Length bias

Liu et al. 2025, Sec. 3.2

In Dr. GRPO:

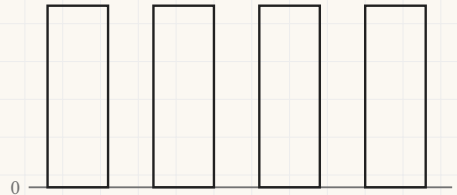
$$\left(r^{(i)} - \bar{r}\right) \times \frac{1}{C}$$

The loop's multiplier, every token alike:
this morning's policy gradient

C: the generation budget, the same for every answer

Groups whose answers all agree teach nothing

Every answer to x correct

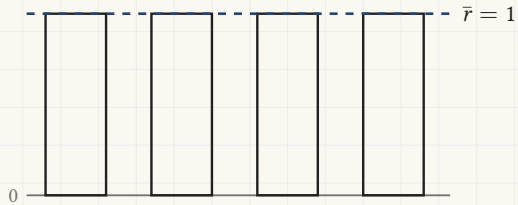


Reward $r^{(i)}$ 1 1 1 1

Groups whose answers all agree teach nothing

Every answer to x correct

Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$

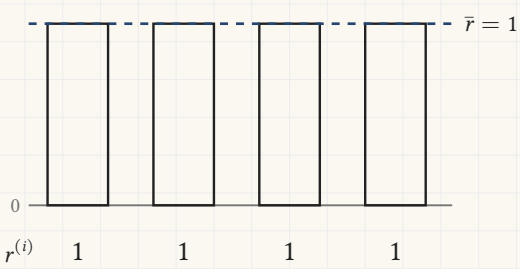


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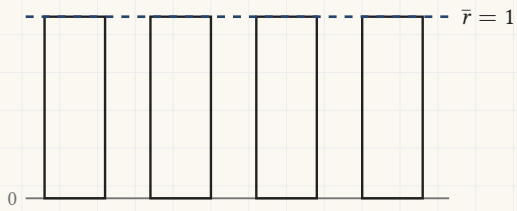


$$\frac{r^{(i)} - \bar{r}}{\text{std}} = \frac{0}{0}$$

Groups whose answers all agree teach nothing

Every answer to x correct

Group average $\bar{r} = 1$: every multiplier $r^{(i)} - \bar{r} = 0$



$$\frac{r^{(i)} - \bar{r}}{\text{std}} = \frac{0}{0}$$

Code adds a small δ:

$$\frac{r^{(i)} - \bar{r}}{\text{std} + \delta} = \frac{0}{\delta} = 0$$

The group adds nothing to the step: its n answers were wasted

Five ways to fill the multiplier

Algorithm	Multiplier	Score	In words
REINFORCE	G_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Rest of the episode
With a baseline	$G_t - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than usual here
Actor-critic	$r_t + \gamma\, V(s_{t+1}) - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than predicted
PPO	Ψ_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Same, step constrained

Five ways to fill the multiplier

Algorithm	Multiplier	Score	In words
REINFORCE	G_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Rest of the episode
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Actor-critic	$r_t + \gamma\, V(s_{t+1}) - V(s_t)$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than predicted
PPO	Ψ_t	$\nabla_w \log \pi_w(a_t \mid s_t)$	Same, step constrained
GRPO	$(r^{(i)} - \bar{r})/\text{std}$	$\nabla_w \log \pi_w(a_t \mid s_t)$	Better than its group

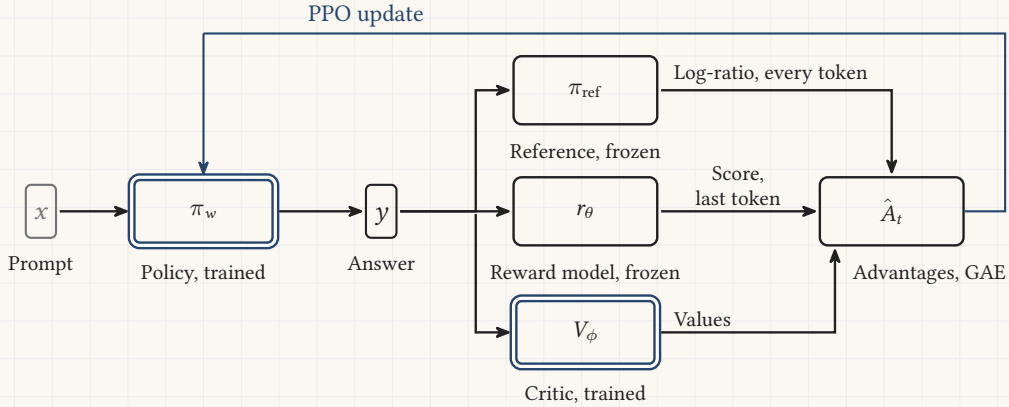
std over n or $n-1$

4

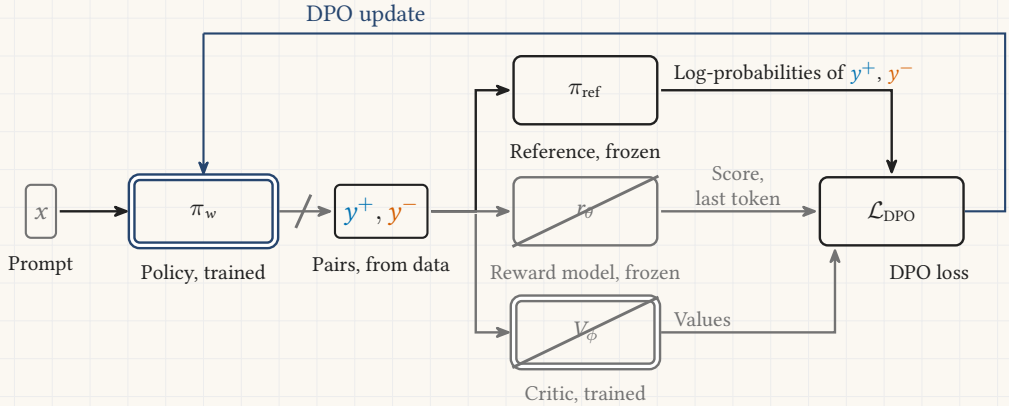
Section 4

Direct preference optimisation

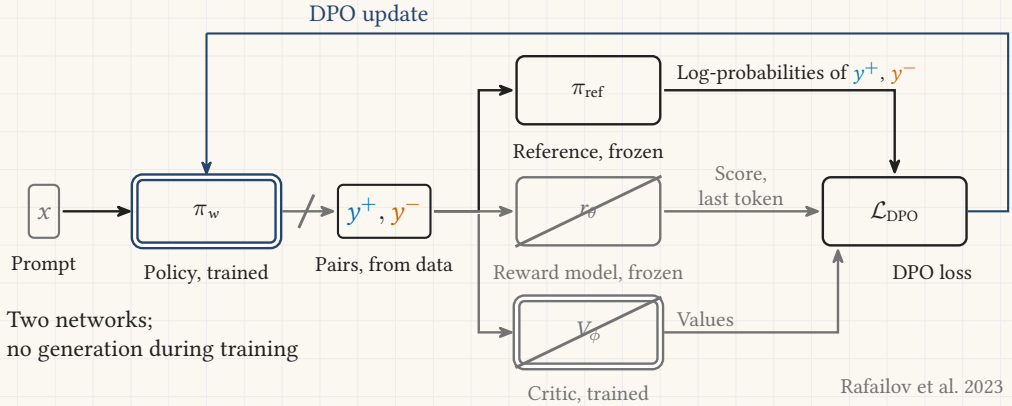
What DPO removes



What DPO removes



What DPO removes

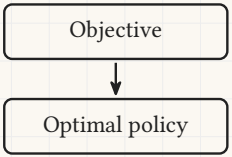


From Part 1's objective to the DPO loss

Objective

$$\mathcal{J}(\pi) = \underset{\text{Answers } y \text{ drawn from any } \pi}{\mathbb{E}}\big[r(x, y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\text{ref}}\big)$$

From Part 1's objective to the DPO loss



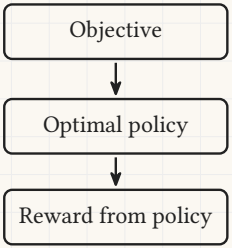
$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

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$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

From Part 1's objective to the DPO loss

Objective

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

Optimal policy

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

Reward from policy

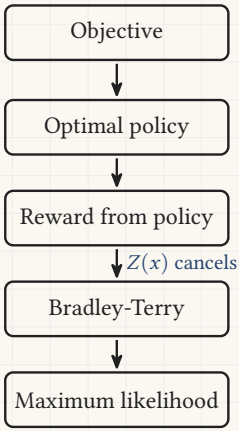
$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$Z(x)$ cancels

Bradley-Terry

$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

From Part 1's objective to the DPO loss



$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

$$\mathcal{L}_{\mathrm{DPO}}(w) = -\mathbb{E}\big[\log P_w(y^+ \succ y^- \mid x)\big]$$

P_w: π_w for π^*

From Part 1's objective to the DPO loss

Objective

$$\mathcal{J}(\pi) = \mathbb{E}\big[r(x,y)\big] - \beta\,\mathrm{KL}\big(\pi \,\big\|\, \pi_{\mathrm{ref}}\big)$$

Answers y drawn from any π

Rafailov et al. 2023, eqs. (3) to (7)

Optimal policy

$$\pi^*(y \mid x) = \frac{\pi_{\mathrm{ref}}(y \mid x)\, e^{r(x,y)/\beta}}{Z(x)}$$

$Z(x)$: a sum over answers

Reward from policy

$$r(x,y) = \beta \log \frac{\pi^*(y \mid x)}{\pi_{\mathrm{ref}}(y \mid x)} + \beta \log Z(x)$$

$Z(x)$ cancels

Bradley-Terry

$$P(y^+ \succ y^- \mid x) = \sigma\left(\beta \log \frac{\pi^*(y^+ \mid x)}{\pi_{\mathrm{ref}}(y^+ \mid x)} - \beta \log \frac{\pi^*(y^- \mid x)}{\pi_{\mathrm{ref}}(y^- \mid x)}\right)$$

Maximum likelihood

$$\mathcal{L}_{\mathrm{DPO}}(w) = -\mathbb{E}\big[\log P_w(y^+ \succ y^- \mid x)\big]$$

P_w: π_w for π^*

The DPO loss

Policy, trained

$$\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)}$$

The DPO loss

Policy, trained

$$\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi_w(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}$$

KL weight

Reference, frozen

The DPO loss

Policy, trained

Sigmoid

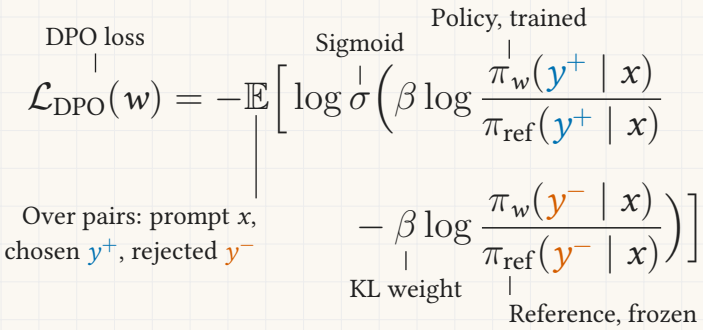
$$\log \sigma\left(\beta \log \frac{\pi_w(y^+ \mid x)}{\pi_{\text{ref}}(y^+ \mid x)} - \beta \log \frac{\pi_w(y^- \mid x)}{\pi_{\text{ref}}(y^- \mid x)}\right)$$

KL weight

Reference, frozen

The DPO loss

Rafailov et al. 2023, eq. (7)



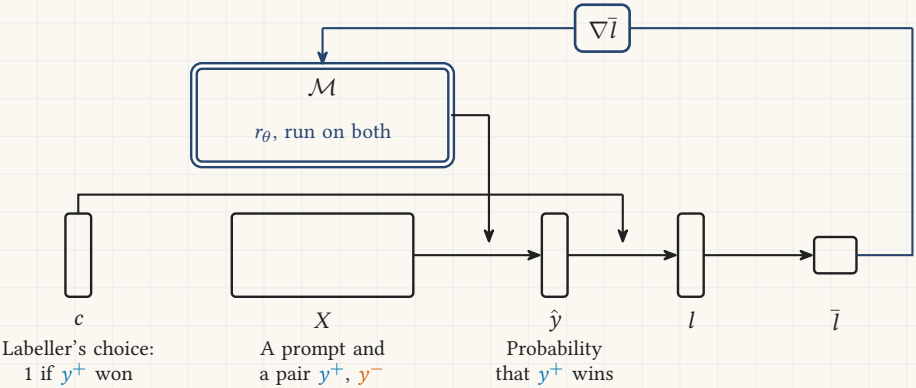
5

Section 5

DPO in practice

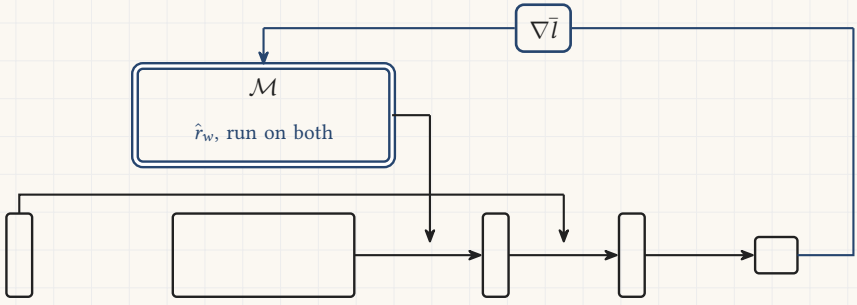
The DPO loss is a reward-model loss

$$\hat{y} = \sigma\big(r_\theta(x, y^+) - r_\theta(x, y^-)\big) \qquad l = -\log \hat{y}$$



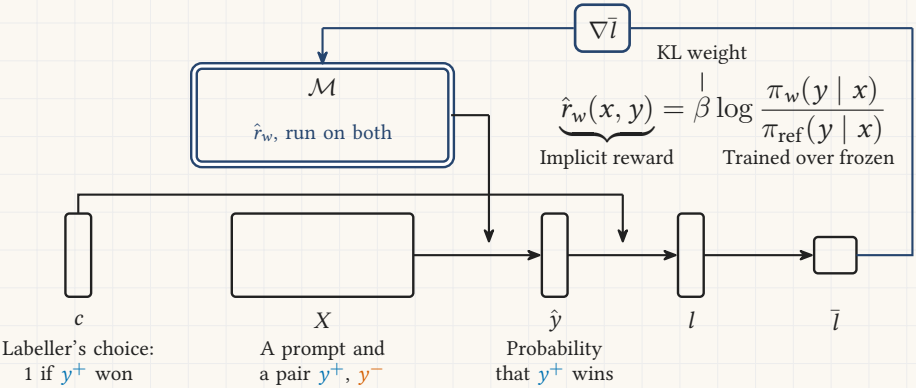
The DPO loss is a reward-model loss

$$\hat{y} = \sigma\left(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\right) \qquad l = -\log \hat{y}$$



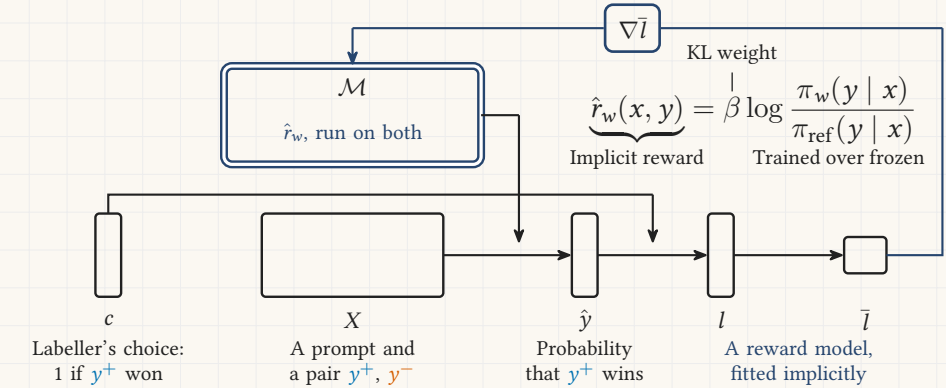
The DPO loss is a reward-model loss

$$\hat{y} = \sigma\left(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\right) \qquad l = -\log \hat{y}$$



The DPO loss is a reward-model loss

$$\hat{y} = \sigma\big(\hat{r}_w(x, y^+) - \hat{r}_w(x, y^-)\big) \qquad l = -\log \hat{y}$$



References

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A

Appendix

The group average as a baseline

The group average includes the answer's own reward

This morning's rule

$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

The group average includes the answer's own reward

This morning's rule

$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

A baseline $b(x)$ may look at the prompt x, never at the answer y

Answer $y^{(i)}$	ollh	ollh	ollh	olleh	
Reward $r^{(i)}$	0	0	0	1	
Group average $\bar{r}$	0.25	0.25	0.25	0.25	Sees $y^{(i)}$: breaks the rule

The group average includes the answer's own reward

This morning's rule

$$\mathbb{E}\big[\, b(x)\, \nabla_w \log \pi_w(y \mid x) \,\big] = b(x)\, \mathbb{E}\big[\nabla_w \log \pi_w(y \mid x)\big] = 0$$

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Answer $y^{(i)}$	ollh	ollh	ollh	olleh	
Reward $r^{(i)}$	0	0	0	1	
Group average $\bar{r}$	0.25	0.25	0.25	0.25	Sees $y^{(i)}$: breaks the rule
Others' average $\bar{r}_{-i}$	0.33	0.33	0.33	0	Never sees $y^{(i)}$: keeps it

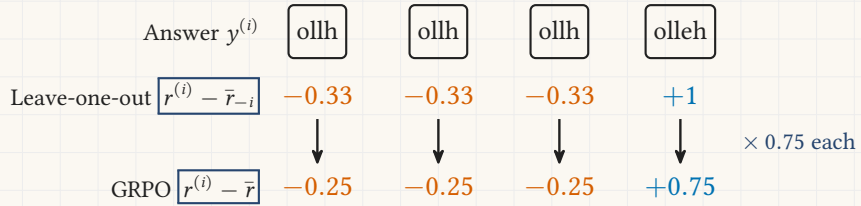
Welleck 2026; Kool et al. 2019; Ahmadian et al. 2024

The group average is the others' average, rescaled

Answer $y^{(i)}$	ollh	ollh	ollh	olleh
Leave-one-out $r^{(i)} - \bar{r}_{-i}$	-0.33	-0.33	-0.33	$+1$
GRPO $r^{(i)} - \bar{r}$	-0.25	-0.25	-0.25	$+0.75$

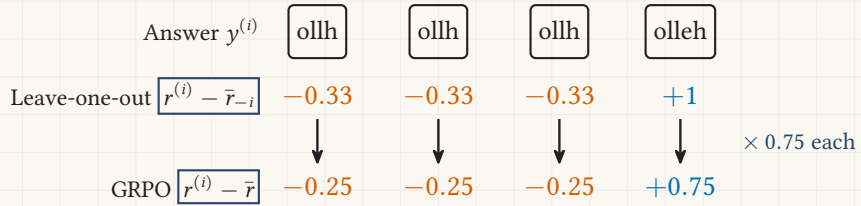
Welleck 2026, CMU CS11-711, lecture 17

The group average is the others' average, rescaled



Welleck 2026, CMU CS11-711, lecture 17

The group average is the others' average, rescaled

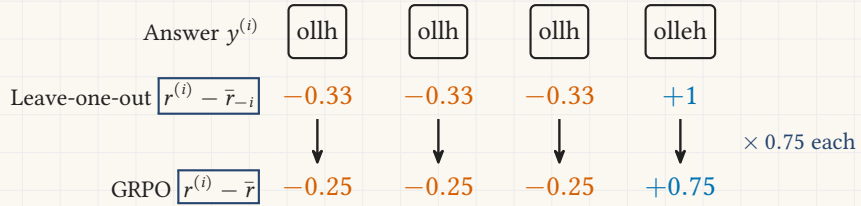


$$r^{(i)} - \bar{r} = \frac{n-1}{n}\left(r^{(i)} - \bar{r}_{-i}\right)$$

since $n\,\bar{r} = r^{(i)} + (n-1)\,\bar{r}_{-i}$

Welleck 2026, CMU CS11-711, lecture 17
Kool et al. 2019; Liu et al. 2025, App. A

The group average is the others' average, rescaled



$$r^{(i)} - \bar{r} = \frac{n-1}{n}\left(r^{(i)} - \bar{r}_{-i}\right)$$

since $n\,\bar{r} = r^{(i)} + (n-1)\,\bar{r}_{-i}$

Leave-one-out: on average the policy gradient
GRPO: times $(n-1)/n$; the step size absorbs it

Welleck 2026, CMU CS11-711, lecture 17
Kool et al. 2019; Liu et al. 2025, App. A